Kernix 0.12.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

152 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

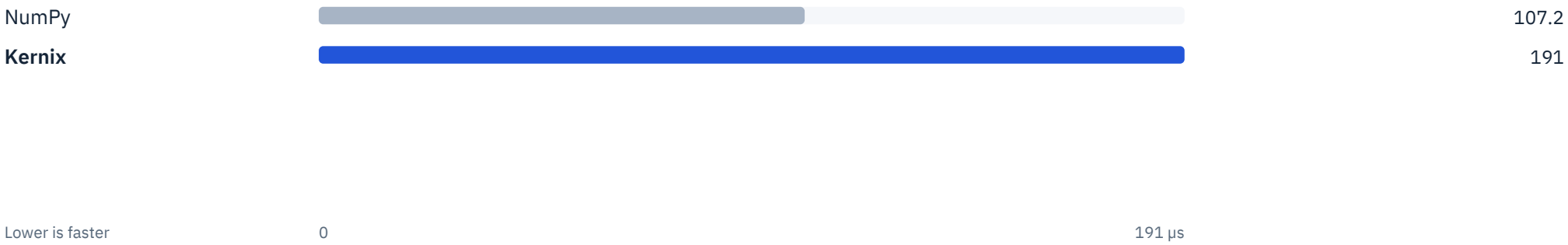
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

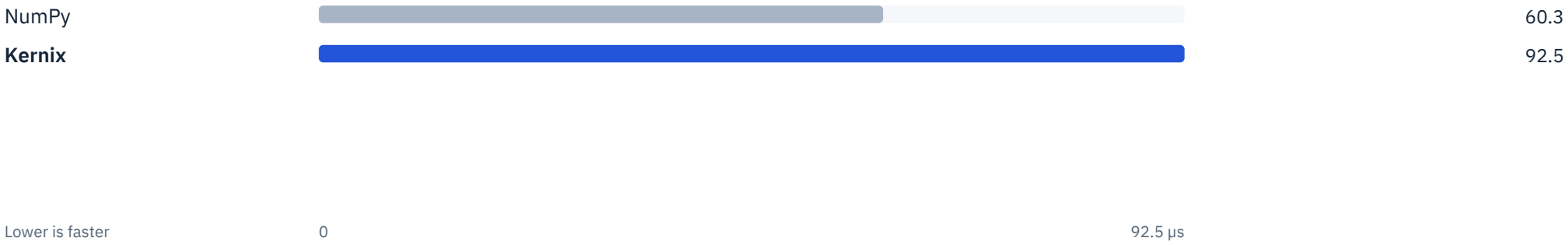
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



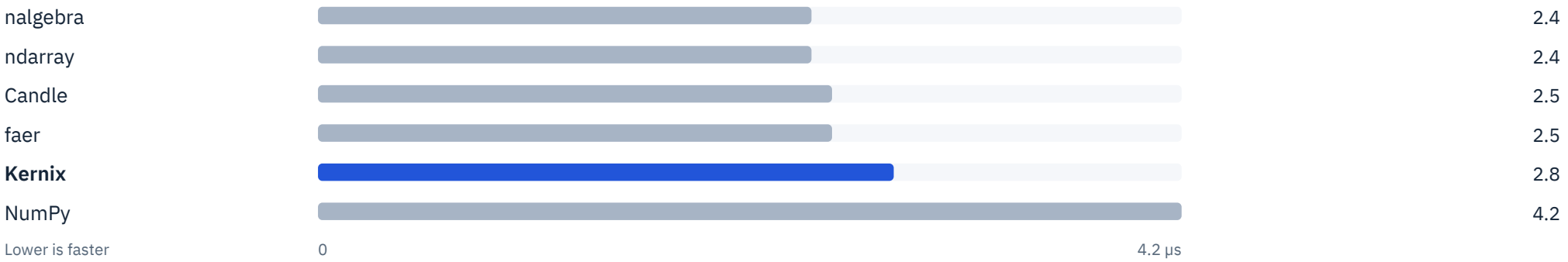
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



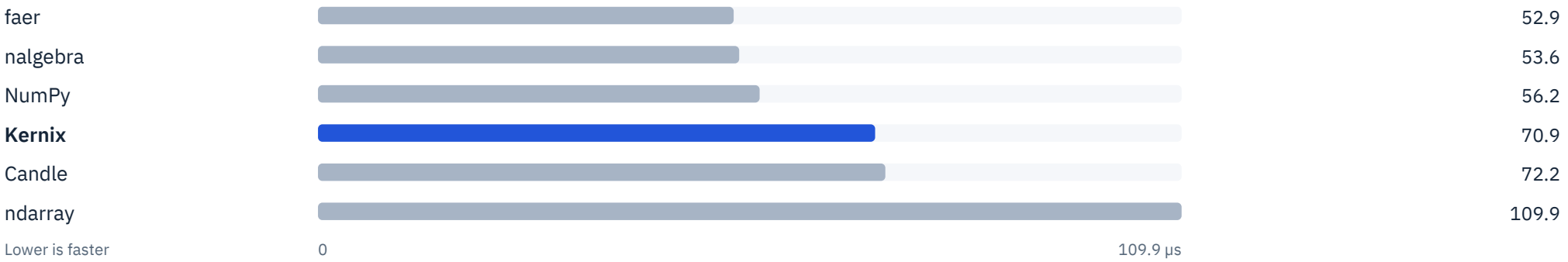
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



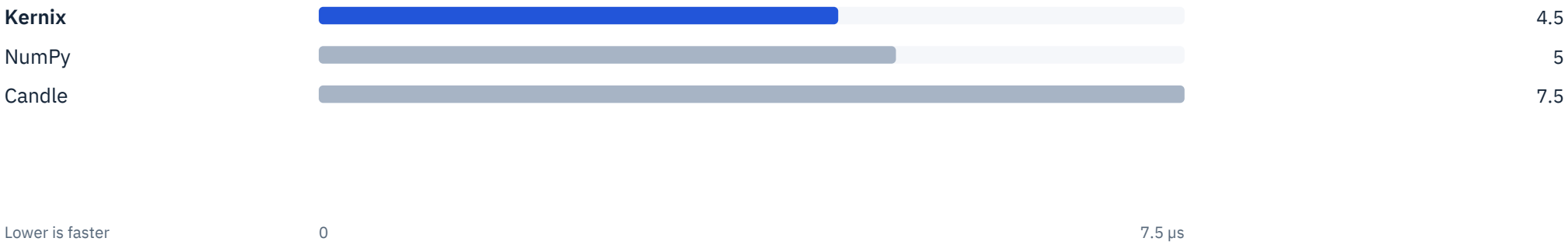
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

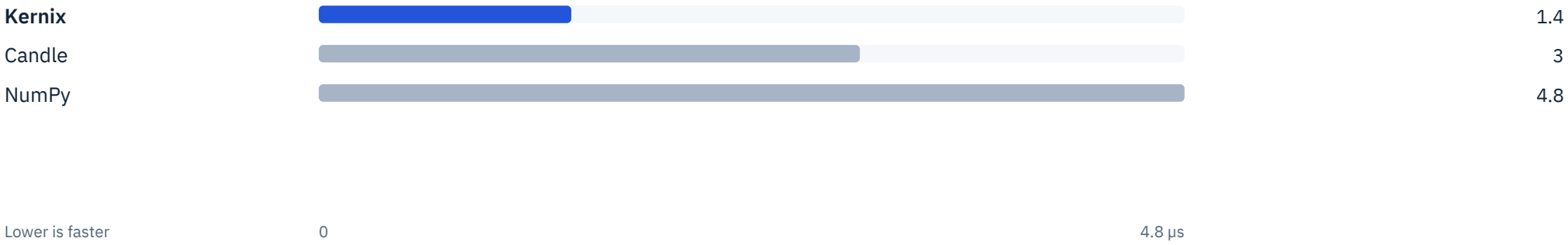
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



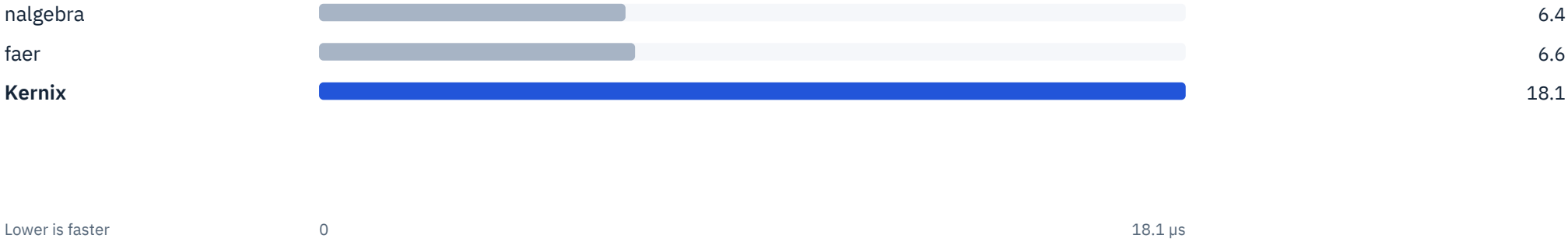
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



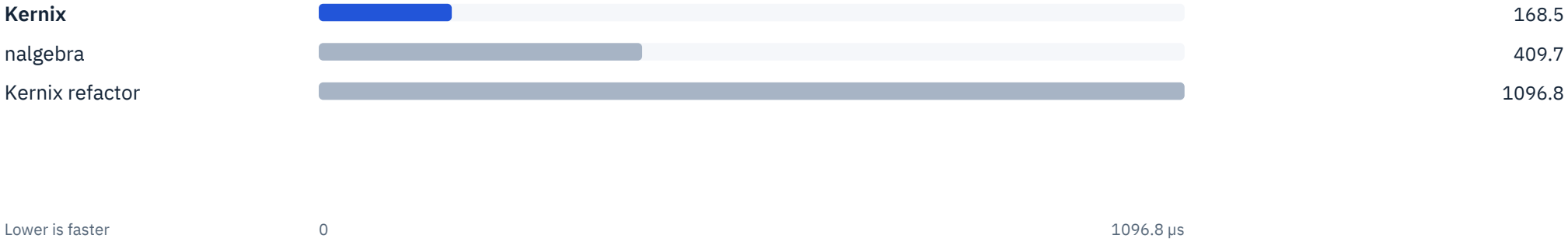
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

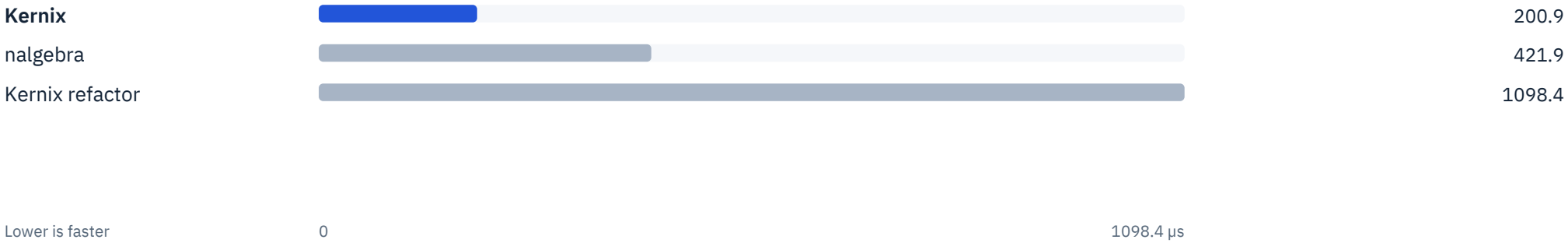
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



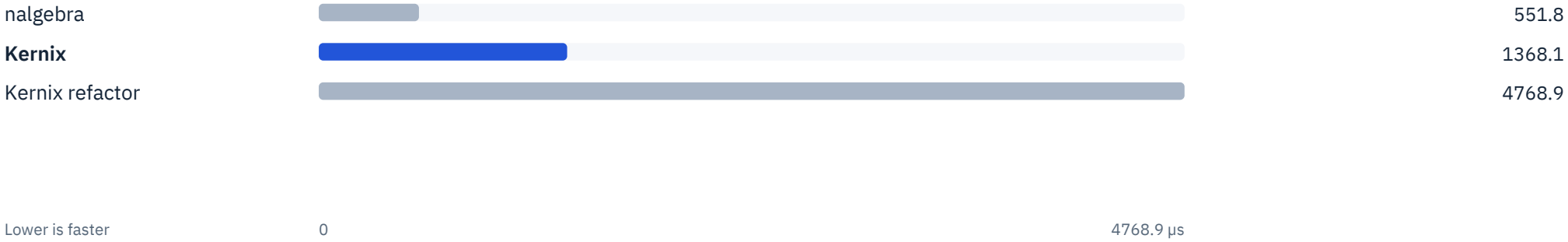
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

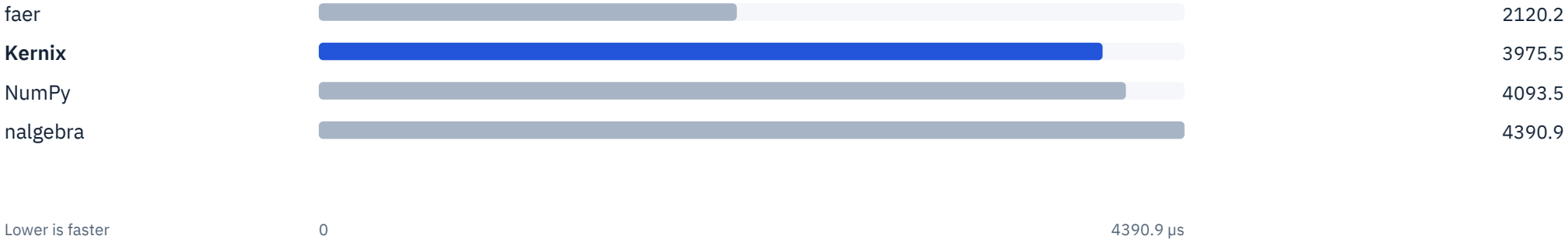
512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output

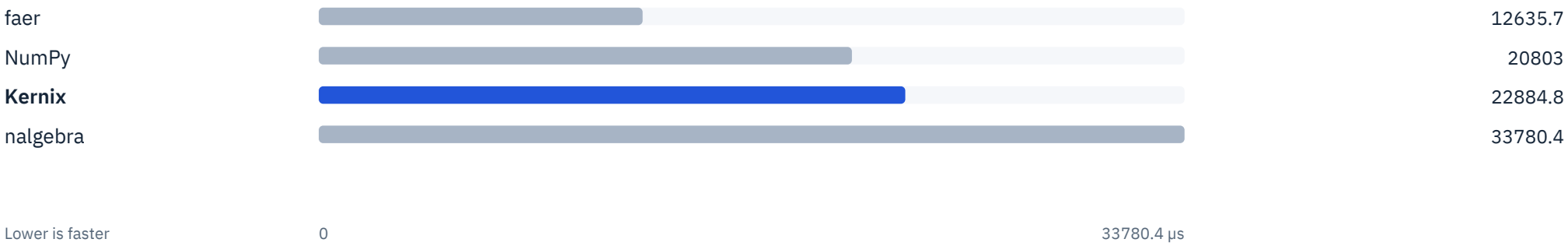


Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

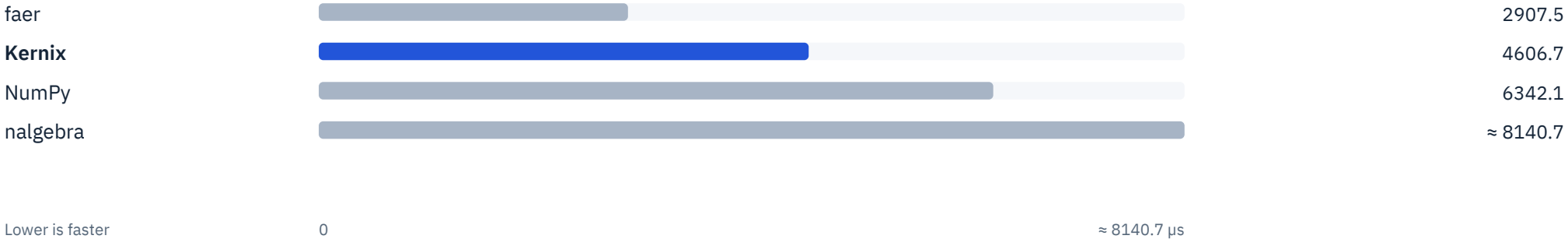
Thin QR

1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

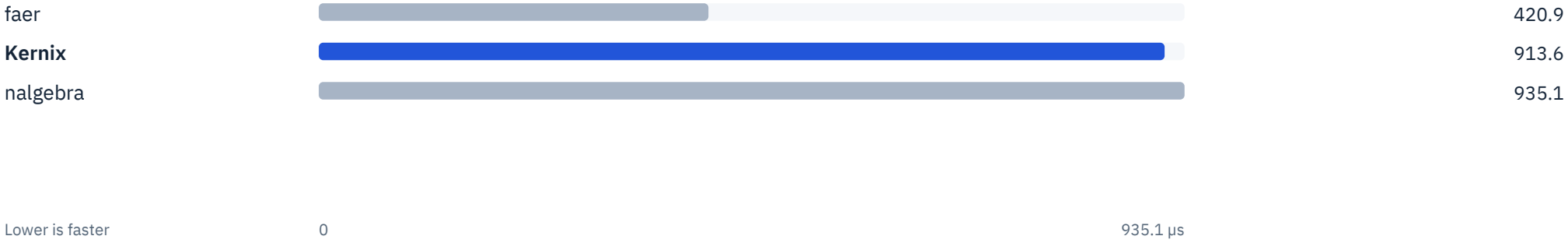
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



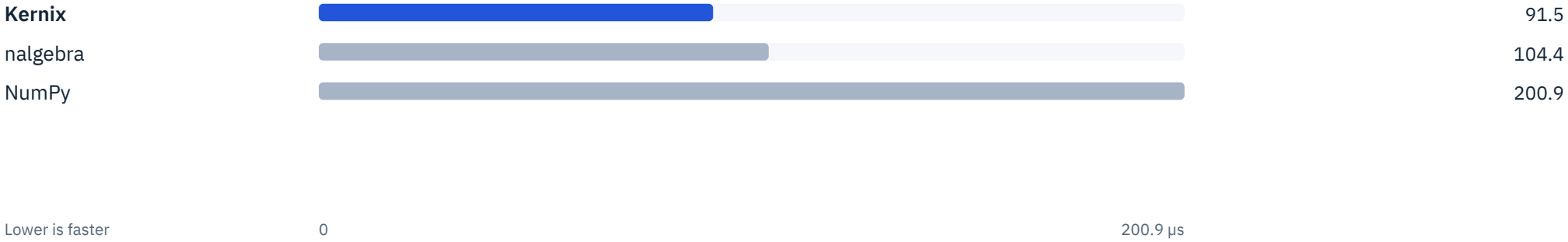
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

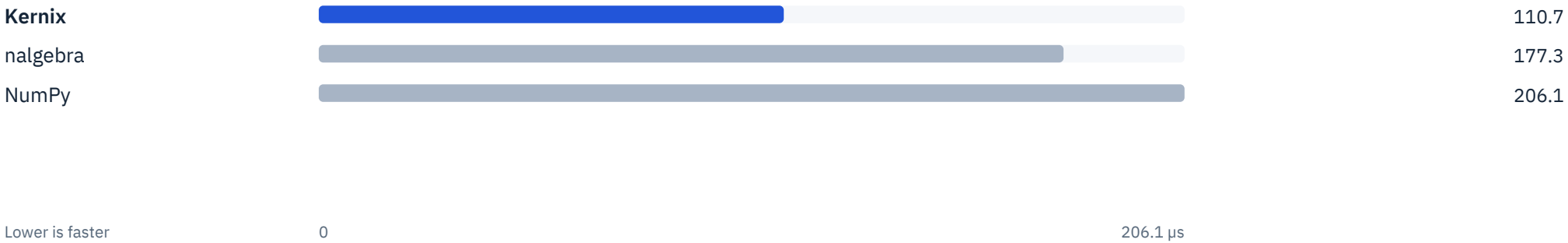
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

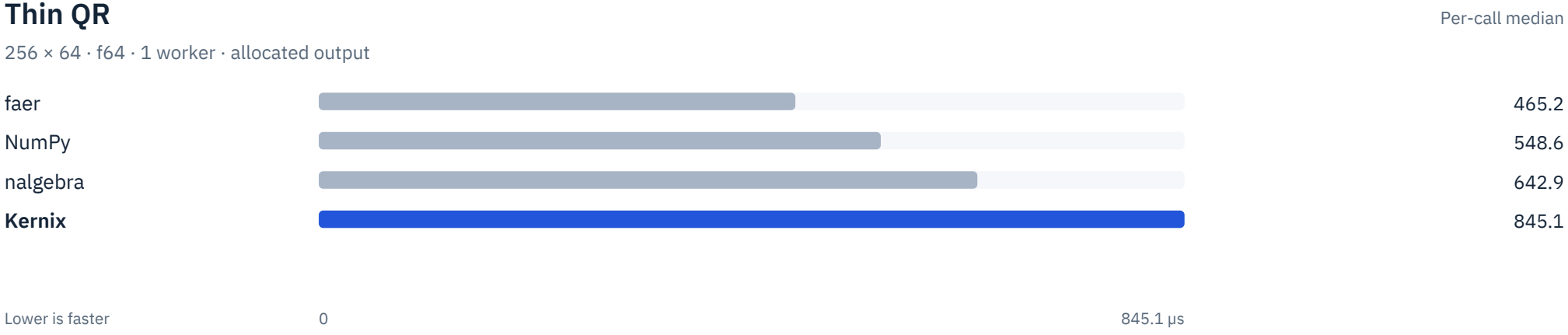
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



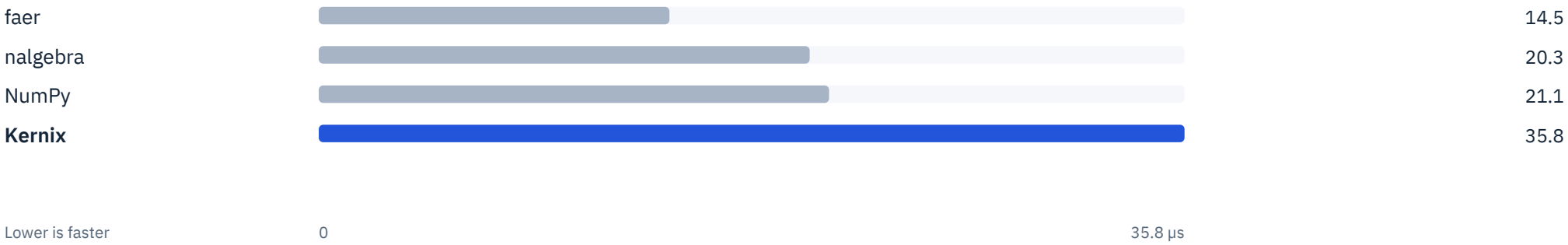
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

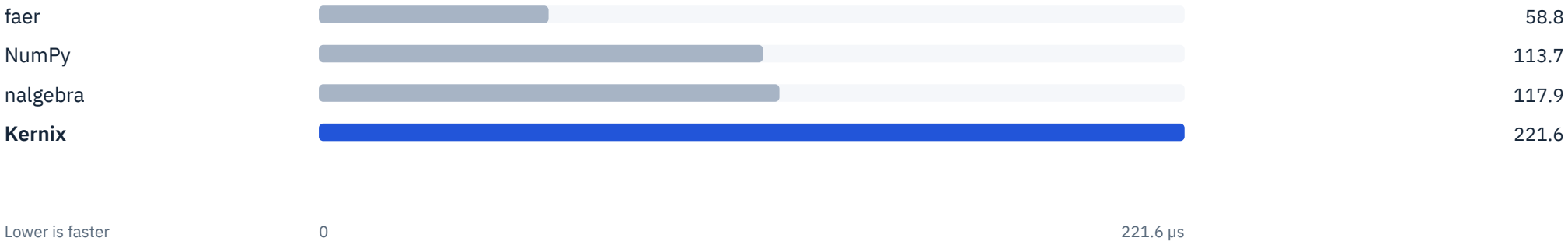
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



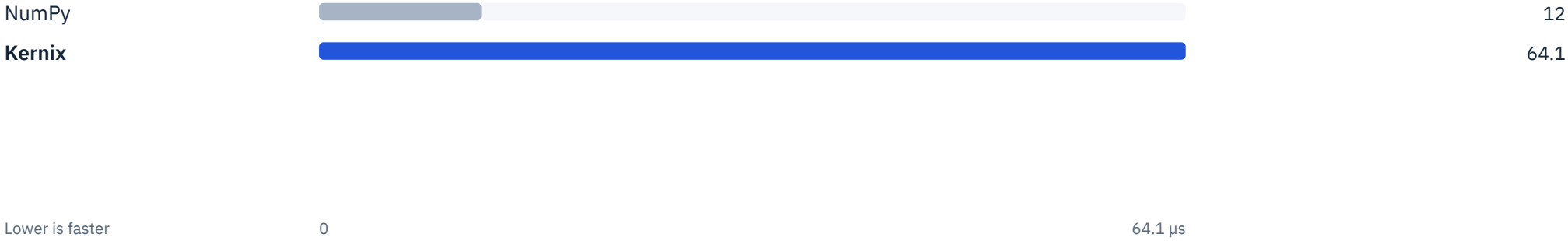
Kernix 0.12.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



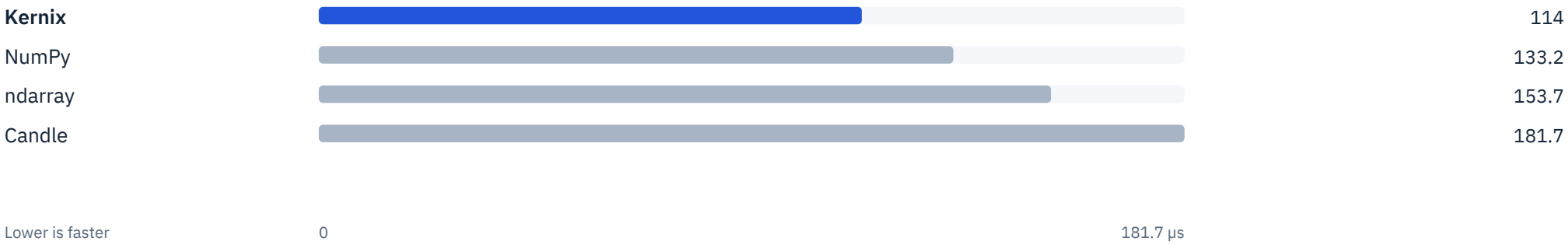
Kernix 0.12.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median



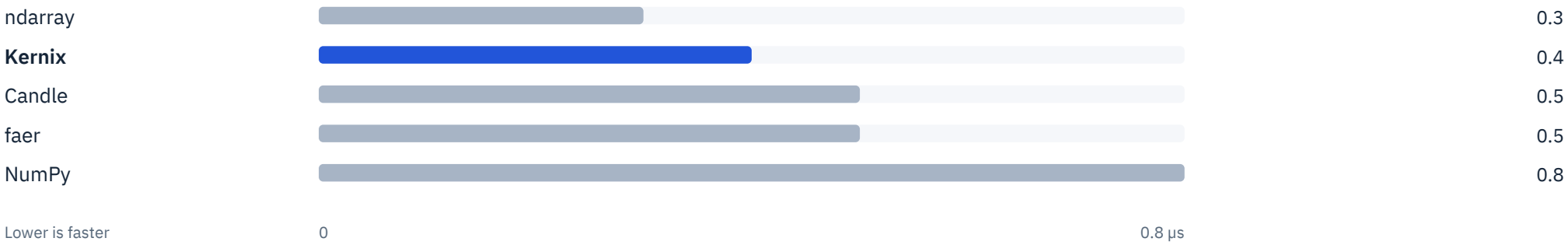
Kernix 0.12.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9738219895287958	0.968421052631579 – 0.9738219895287958

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.18717663909268947	0.1865605658709107 – 0.18858212447356107

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.22232916265640038	0.22212072115550346 – 0.22289301685586516

Thin QR

512 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7779124044095739	0.7753794875913581 – 0.7799979937807202

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.578889476761995	0.5775322067599985 – 0.5826379763927365

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8276896699528504	0.8116910229645093 – 0.8392311052861512

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8266666666666667	0.8089117395029991 – 0.8577699736611062

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9572822465492623	0.9372616558002215 – 1.021105141980046

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9989429175475687	0.6957478005865103 – 1.4261603375527425

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.6532593619972261 – 1.0021276595744681

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	**0.9964324917672887**	95% INTERVAL	0.8442432683379758 – 1.0080343537886134

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9974604966139955	0.843441466854725 – 1.0025359256128488

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	0.9791666666666666 – 1.0208333333333333

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9978308026030369	0.9956772334293948 – 1.032490974729242

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9919571045576407	0.908344733242134 – 1.0027359781121752

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	1 – 1.28

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	1 – 1

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0008684324793746	0.9608865710560626 – 1.0411206507004067

Cholesky factorization

256 × 256 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0045150014564521	1.0007302468234263 – 1.0139275766016713

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0013484710078733	0.979931106784484 – 1.0090029357399153

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0096153846153846	0.9894490035169988 – 1.0169491525423728

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0204508482454102	0.7212323064113239 – 1.3266775017413512

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0017421602787457	0.9702797202797203 – 1.2707993474714518

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	191000	191
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	107200	107.2
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	92500	92.5
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	60300	60.3
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	136700	136.7
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101100	101.1
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	71500	71.5
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56700	56.7
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4200	4.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	52700	52.7
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14100	14.1
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	52100	52.1
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15900	15.9
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13600	13.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	136100	136.1
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	113800	113.8
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	111000	111
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	139100	139.1
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	117400	117.4
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100900	100.9
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1132300	1132.3
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	956200	956.2
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	756800	756.8
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1144800	1144.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	855400	855.4
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	855400	855.4
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6900	6.9
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10400	10.4
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7200	7.2
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7800	7.8
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	26100	26.1
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	70900	70.9
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	109900	109.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	52900	52.9
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53600	53.6
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56200	56.2
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	72200	72.2
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	394100	394.1
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	425100	425.1
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	421900	421.9
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	386500	386.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	389100	389.1
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	451100	451.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	4500	4.5
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	7500	7.5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1400	1.4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	4800	4.8
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	3000	3
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	39700	39.7
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	96700	96.7
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	446600	446.6
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	60900	60.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	6600	6.6
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	6400	6.4
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18100	18.1
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1282600	1282.6
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1390700	1390.7
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	324500	324.5
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	9497700	9497.7
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2268500	2268.5
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	10940400	10940.4
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1196900	1196.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1289500	1289.5
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	362900	362.9
Cholesky rank-one update / nalgebra 256 × 256 · f64	409700	409.7
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	1096800	1096.8
Cholesky rank-one update / Kernix 256 × 256 · f64	168500	168.5
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	421900	421.9
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	1098400	1098.4
Cholesky rank-one downdate / Kernix 256 × 256 · f64	200900	200.9
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4768900	4768.9
Cholesky rank-one update / nalgebra 512 × 512 · f64	551800	551.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix 512 × 512 · f64	1368100	1368.1
Thin QR / Kernix 512 × 128 · f64	3975500	3975.5
Thin QR / nalgebra 512 × 128 · f64	4390900	4390.9
Thin QR / NumPy 512 × 128 · f64	4093500	4093.5
Thin QR / faer 512 × 128 · f64	2120200	2120.2
Thin QR / faer 1024 × 256 · f64	12635700	12635.7
Thin QR / NumPy 1024 × 256 · f64	20803000	20803
Thin QR / nalgebra 1024 × 256 · f64	33780400	33780.4
Thin QR / Kernix 1024 × 256 · f64	22884800	22884.8
Cholesky factorization / NumPy 512 × 512 · f64	6342100	6342.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	8140700.000000001	8140.700000000001
Cholesky factorization / faer 512 × 512 · f64	2907500	2907.5
Cholesky factorization / Kernix 512 × 512 · f64	4606700	4606.7
Column-pivoted QR / faer 96 × 32 · f64	69000	69
Column-pivoted QR / Kernix 96 × 32 · f64	137700	137.7
Column-pivoted QR / nalgebra 96 × 32 · f64	105500	105.5
Column-pivoted QR / nalgebra 256 × 64 · f64	935100	935.1
Column-pivoted QR / Kernix 256 × 64 · f64	913600	913.6
Column-pivoted QR / faer 256 × 64 · f64	420900	420.9
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	91500	91.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	104400	104.4
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	200900	200.9
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	177300	177.3
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	110700	110.7
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	206100	206.1
Thin QR / nalgebra 96 × 32 · f64	79300	79.3
Thin QR / NumPy 96 × 32 · f64	127200	127.2
Thin QR / faer 96 × 32 · f64	83000	83
Thin QR / Kernix 96 × 32 · f64	111100	111.1
Thin QR / NumPy 256 × 64 · f64	548600	548.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	642900	642.9
Thin QR / faer 256 × 64 · f64	465200	465.2
Thin QR / Kernix 256 × 64 · f64	845100	845.1
Cholesky factorization / faer 64 × 64 · f64	14500	14.5
Cholesky factorization / Kernix 64 × 64 · f64	35800	35.8
Cholesky factorization / nalgebra 64 × 64 · f64	20300	20.3
Cholesky factorization / NumPy 64 × 64 · f64	21100	21.1
Cholesky factorization / faer 128 × 128 · f64	58800	58.8
Cholesky factorization / NumPy 128 × 128 · f64	113700	113.7
Cholesky factorization / Kernix 128 × 128 · f64	221600	221.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 128 × 128 · f64	117900	117.9
Cholesky factorization / Kernix 256 × 256 · f64	690100	690.1
Cholesky factorization / nalgebra 256 × 256 · f64	869800	869.8
Cholesky factorization / NumPy 256 × 256 · f64	937800	937.8
Cholesky factorization / faer 256 × 256 · f64	317800	317.8
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	58000	58
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10500	10.5
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9800	9.8
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4200	4.2
Model inference / Candle batched normalized/sliced MLP · f64	15400.000000000002	15.400000000000002

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4900	4.9
Model recording + forward / Candle batched normalized/sliced MLP · f64	10200	10.2
Model backward / Kernix prepared graph → fresh gradients · f64	5900	5.9
Model backward / Candle prepared graph → fresh gradients · f64	32000	32
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	64100	64.1
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	12000	12
Row softmax / Kernix 256 × 64 · f64	114000	114
Row softmax / ndarray 256 × 64 · f64	153700	153.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	133200	133.2
Row softmax / Candle 256 × 64 · f64	181700	181.7
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create zeros / faer 64 × 64 · f32	500	0.5
Create zeros / Candle 64 × 64 · f32	500	0.5
Create ones / faer 64 × 64 · f32	500	0.5
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / NumPy 64 × 64 · f32	2100	2.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / Kernix 64 × 64 · f32	500	0.5